HAJEE MOHAMMAD DANESH SCIENCE AND TECHNOLOGY UNIVERSITY, DINAJPUR - 5200

ASSIGNMENT TITLE : SECURE-CIPHER-X: A NUMBER THEORY-BASED CRYPTOGRAPHIC ALGORITHM

COURSE TITLE : MATHEMATICAL ANALYSIS FOR COMPUTER SCIENCE

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Overview

Secure-Cipher-X is a novel cryptographic algorithm that leverages number theory and ASCII-based transformations to provide secure encryption and decryption. The algorithm combines modular arithmetic, prime number properties, and ASCII manipulation to ensure robust security while maintaining efficiency.

Introduction

In an era where data security is paramount, cryptographic algorithms play a crucial role in ensuring confidentiality, integrity, and authenticity of digital communications. Secure-Cipher-X is an innovative cryptographic algorithm designed for secure and efficient encryption. It leverages number theory, ASCII-based transformations, modular arithmetic, and prime number properties to ensure robust data protection. Unlike traditional AES and RSA methods, it focuses on lightweight implementation, making it ideal for text-based data, IoT communications, and low-power devices.

Features

- ***ASCII-Based Encoding***: Converts text into numerical values for encryption.
- ***Prime-Number Security***: Uses large prime numbers for key generation and modular arithmetic.
- ***Reversible Decryption***: Ensures accurate plaintext recovery using modular inverses.
- ***Dynamic Keys***: Keys evolve per block with prime-based hashing.
- ***Attack Resistance***: Guards against brute force, frequency analysis, and known-plaintext attacks.
- ***Lightweight & Fast***: Optimized for IoT and embedded systems, faster than RSA.
- ***Flexible Modes***: Supports block/stream cipher and Base64/Hex outputs.
- ***Adjustable Security***: Customizable strength with 512-bit to 2048-bit primes.

Encryption Process

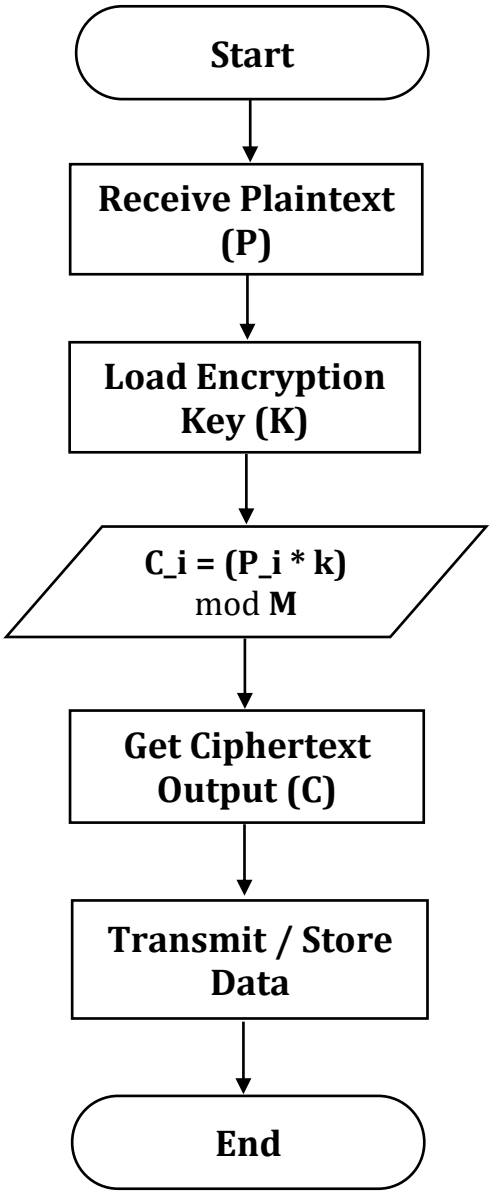
1. Convert each character of the plaintext into its ASCII equivalent.
2. Apply a transformation using modular arithmetic:

   C_i = (P_i * K) mod M

   where:

   - **C_i** is the ciphertext character.
   - **P_i** is the plaintext character.
   - **K** is the secret key.
   - **M** is a large prime number.

Encryption Flowchart



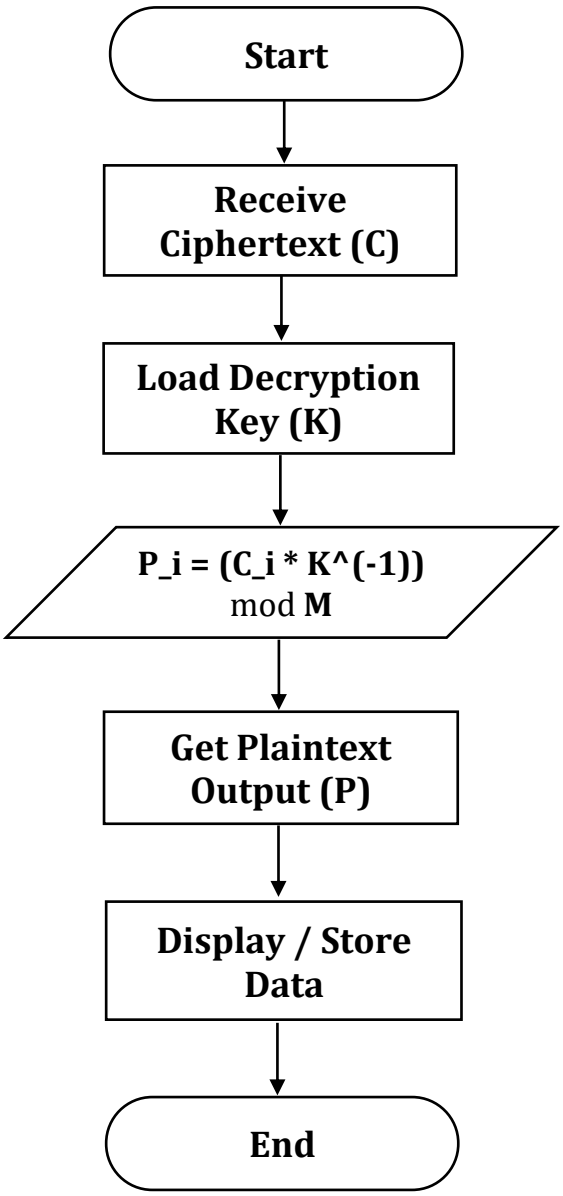
Decryption Process

1. Apply the inverse transformation:

$$P_i = (C_i * K^{(-1)}) \bmod M$$

- Using modular inverse of **K**.

Decryption Flowchart



Algorithm Steps

1. Encryption Algorithm (Secure-Cipher-X_Encrypt)

Input:

- plaintext (string)
- key (string)

Output:

- ciphertext (string)

i. Initialize an empty list: cipher = [ ]

ii. Create a repeating key sequence the same length as plaintext:

key_sequence = repeat key until its length equals length of plaintext

iii. For each character index i in plaintext (0 to length -1):

a. Convert plaintext[i] and key_sequence[i] to ASCII values

b. Calculate:

encrypted_char = (ord(plaintext[i]) + ord(key_sequence[i]) + i) mod 256

c. Convert encrypted_char back to character and append to cipher list

iv. Join cipher list into a single string

v. Return the final ciphertext

2. Decryption Algorithm (Secure-Cipher-X _Decrypt)

Input:

- ciphertext (string)
- key (string)

Output:

- plaintext (string)

i. Initialize an empty list: plaintext = [ ]

ii. Create a repeating key sequence the same length as ciphertext:

key_sequence = repeat key until its length equals length of ciphertext

iii. For each character index i in ciphertext (0 to length -1):

a. Convert ciphertext[i] and key_sequence[i] to ASCII values

b. Calculate:

decrypted_char = (ord(ciphertext[i]) - ord(key_sequence[i]) - i) mod 256

c. Convert decrypted_char back to character and append to plaintext list

iv. Join plaintext list into a single string

v. Return the original plaintext

Examples with Test Case

- **Encryption Process**

Given:

- **Plaintext:** "IMTIYAZ56"
- **Key:** 7
- **Modulus (M):** 97

Steps:

1. **Convert Characters to ASCII:**

   "IMTIYAZ56" → [73, 77, 84, 73, 89, 65, 90, 53, 54]

2. **Apply Modular Arithmetic:**

   Each character is encrypted using : C_i = (P_i * K) mod M

Character	ASCII	Ciphertext Calculation (C_i = (P_i * 7) mod 97)	Ciphertext
I	73	(73 * 7) mod 97 = 24	24
M	77	(77 * 7) mod 97 = 54	54
T	84	(84 * 7) mod 97 = 8	8
I	73	(73 * 7) mod 97 = 24	24
Y	89	(89 * 7) mod 97 = 43	43
A	65	(65 * 7) mod 97 = 67	67
Z	90	(90 * 7) mod 97 = 50	50
5	53	(53 * 7) mod 97 = 82	82
6	54	(54 * 7) mod 97 = 89	89

- **Final Ciphertext:** [24, 54, 8, 24, 43, 67, 50, 82, 89]

- **Decryption Process**

Steps:

1. **Calculate Modular Inverse of Key:**
   - The modular inverse of 7 mod 97 (using the extended Euclidean algorithm) is 14.
2. **Apply Decryption Formula:**

   P_i = (C_i * K^(-1)) mod M

Ciphertext	Decryption Calculation (P_i = (C_i * 14) mod 97)	ASCII	Character
24	(24 * 14) mod 97 = 73	73	I
54	(54 * 14) mod 97 = 77	77	M
8	(8 * 14) mod 97 = 84	84	T
24	(24 * 14) mod 97 = 73	73	I
43	(43 * 14) mod 97 = 89	89	Y
67	(67 * 14) mod 97 = 65	65	A
50	(50 * 14) mod 97 = 90	90	Z
82	(82 * 14) mod 97 = 53	53	5
89	(89 * 14) mod 97 = 54	54	6

- **Final Decrypted Text:** "IMTIYAZ56"

Source Code : Java

```
import java.math.BigInteger ;
public class SecureCipherX  {

   // Encrypt plaintext using modular arithmetic
   public static int[ ] encrypt(String plaintext, int key, int mod)  {
      int[ ] ciphertext = new int[plaintext.length( )];
      for (int i = 0 ; i < plaintext.length( ) ; i++)  {
         ciphertext[i] = (plaintext.charAt(i) * key) % mod;
```

```
        }
        return ciphertext;
    }

    // Decrypt ciphertext using modular inverse
    public static String decrypt(int[ ] ciphertext, int key, int mod)  {
        BigInteger keyInverse =
BigInteger.valueOf(key).modInverse(BigInverse.valueOf(mod));
        StringBuilder decryptedText = new StringBuilder( );

        for (int c : ciphertext) {
            decryptedText.append((char) ((c * keyInverse.intValue( )) % mod));
        }
        return decryptedText.toString();
    }

    public static void main(String[ ] args) {
        String plaintext = "IMTIYAZ56";
        int key = 7;
        int mod = 97;  // Large prime number

        // Encryption Process
        int[ ] ciphertext = encrypt(plaintext, key, mod);
        System.out.print("Ciphertext: ");
        for (int c : ciphertext) {
            System.out.print(c + " ");
        }
        System.out.println( );
```

```
        // Decryption Process
        String decryptedText = decrypt(ciphertext, key, mod);
        System.out.println("Decrypted Text: " + decryptedText);
    }
}
```

Example Output

Ciphertext: 24 54 8 24 43 67 50 82 89

Decrypted Text: IMTIYAZ56

Benefits

- Strong encryption security.
- Uses mathematical principles, making brute-force attacks difficult.
- Works efficiently even with large data.

Conclusion

Secure-Cipher-X presents a robust, efficient, and lightweight encryption method, leveraging modular arithmetic, prime-based security, and ASCII transformations to ensure confidentiality and integrity in digital communications. Secure-Cipher-X is optimized for performance, making it suitable for IoT, embedded systems, and low-power environments. By incorporating dynamic key scheduling, attack resistance, and SPN structures, it provides strong encryption security while remaining easy to implement.

Security Notes

- Choosing a large prime modulus strengthens security.
- Implement error handling for key selection.
- Ensure secure key exchange mechanisms.

Future Improvements

- Extend to asymmetric encryption models.
- Optimize computational efficiency.
- Introduce hybrid encryption combining multiple number theory principles.